

Experiment and Data Analysis Workflow: Photokinesis

Project: Group Management (INFIBIO)

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1. Overview

In this activity, we will use a simple portable microscope to record the motility of microalgae loaded in a quasi-2D chamber. We will observe motility under far-red background illumination and compare it to the case in which the cells are also uniformly illuminated from the side by a UV LED. The cells will respond to UV light by changing their swimming speed in a non-directional manner, a process called photokinesis. We will introduce UV light at varying intensities, following dark adaptation between two consecutive intensities. Finally, we will compare photokinetic behavior at different UV-light intensities and motility during the periods between UV-light stimulation.

Outline of the session

1. Presentation of the session activity and objectives
2. Assemble the microscope and calibrate the images with a stage micrometer.
3. Prepare sample 1 and use it to select imaging parameters for the movies.
4. Prepare sample 2 (if needed) and use it to take the 6 movies: *01_control*, *02_UV-light_I1*, *03_control*, *04_UV_light_I2*, *05_control*, *06_UV_light_I2*
5. Transfer the data and analyse the movies

Note: For each step, it might be useful to read the whole set of instructions before you start to work.

2. UV LED placement, microscope assembly and image calibration

We will briefly review the microscope assembly and discuss how it works.

- **Wear gloves** to avoid leaving fingerprints on the optics should you accidentally touch them.
- Place and adjust the UV LED to uniformly illuminate the sample stage (where the sample will be placed).
- Decide on three illumination intensities for the UV light LED. At the simplest level, choosing just two intensities -a low and a high level should give you a measurable difference in behaviour. The LED's intensity can be controlled using the knob on the LED driver. Mark the knob positions on the cube using a piece of tape and a marker. These marks will then help you identify intensity during the experiment.
- Open the camera software (uEye). It should already be installed on the computer we will use to record the movies. For more information on the software and the camera, see the section "Basics of Microscope Assembly".
- To calibrate the images, we will use the Stage Micrometer [R1L1S5P](#) from Thorlabs: 20 mm Horizontal Scale with 100 μm Divisions and Crosshair. Snap an image of the micrometer. This can then be used, e.g., with Fiji, to measure the separation in pixels between the farthest divisions you can see in the image. From this, you can then estimate the pixel-to-microns conversion of your system.

3. Prepare sample 1 and use it to select the imaging parameters for the movies

We now aim to make a simple sample chamber and use it to determine a set of reasonable parameters for the movies. In particular, we will focus on: background illumination intensity, frame rate, camera exposure, and cell concentration.

- We will make a quasi-2-D chamber with a support glass, called a microscope slide, and a thinner glass, called a coverslip. These will sandwich a spacer made of a thin PDMS sheet with an approximate thickness of 80 μm and a circular hole of a diameter of 5 mm.
- **Wear gloves to avoid greasing the glass by touching it directly with your skin** (skin has a natural layer of grease on top).
- A layer of thin PDMS sheet is spin-coated (to achieve a particular thickness- see the details in How Tos handout) on a plastic petri dish. Make a 5mm hole on this sheet with a biopsy puncher (5 mm), then cut a square of around 22 x 22 mm around this hole, and peel off the square piece. This gives a thin, square PMDS sheet with a 5mm hole. Stick the square sheet onto a microscope slide.
- Using a pipette, place a small droplet (1.2 μL) of cell suspension in the middle of the circular well formed by the PDMS sheet, and top it off with a 22 $\times$ 22 mm coverslip.
- Place the sample on the microscope. Adjust the background illumination, focus position, and camera parameters to be able to get a good, smooth video of the cells.
- We will need to be able to track cells and determine their instantaneous velocity. Discuss whether you might need to dilute the cell suspension to be able to do that.
- If needed, dilute the cell suspension by the required amount, using the cell culture medium (TAP). You can find the recipe [here](#) and [here](#). This diluted suspension will then be used to prepare the sample again.

Question: Are there any cells sticking to the glass surfaces? Is this desirable? What could be done to prevent it? Let's discuss it.

4. Record the movies to be analysed

We want to set on the microscope a sample at a suitable concentration with good illumination and imaging parameters and then record movies of cell behaviour with/without an external blue-light stimulus. These will then be analysed to quantify some aspects of light-induced changes in behaviour.

- Make sure that the UV LED is OFF.
- Keep a stopwatch (e.g., from your mobile or the PC where you're acquiring images) running in the background the whole time and record the time reading at the beginning of each movie.
- If needed, prepare a second sample as mentioned in Sec 3.
- Place the sample on the microscope and bring it into focus. Once ready, cover the setup with the black cloth. We will use this to prevent spurious light stimuli to the cells.
- Set the camera to the imaging parameters that you chose in Step 3 and record a 15s movie. Save it as *01_control*.
- Prepare the camera for another 15-second movie. Turn ON the UV LED for 2s at the first light intensity and immediately start recording the movie. Once the recording is over, turn OFF the UV LED. Save the movie as *02_UV_light_I1*.
- With the same protocol, record another movie with the UV LED off (*03_control*), a second movie with light at the other intensity (*04_UV_light_I2*).

- Repeat until the desired light intensity (or until the light intensities you decided to perform the experiment with).
- The experimental part of the activity is done. You can dispose of the sample and cells.

5. Transfer and analyse the movies

Each student will now use their laptop to analyse one of the movies. This minimises the time needed for tracking and maximises the hands-on part (of course the idea is to then share the results!). We will use Matlab and track the cell movement with a set of codes that are widely used in the field (there are plenty of other options, of course, see for example [TrackPy](#) for Python).

Tracking here follows two steps. The first is called "featuring" and involves finding the features in each frame (their coordinates, their integrated intensity and their size). The second step is called "tracking", which means assigning labels to features across the frames, in such a way that the same object is labelled in the same way throughout its trajectory. This allows us to pull out its trajectory.

We will discuss this in a bit more depth during the exercise, but you can also find a in-depth recap of this tracking procedure in this [handout](#)

 **Particle_tracking_handout.pdf**

The required codes are in the GitHub repository, together with some wrapper scripts. [The scripts are the ones we will manipulate.](#)

Our analysis will follow this set of steps:

1. Calculate the median frame from the original frames of the movie and generate a set of median-subtracted frames. These are the ones that will be used for tracking. Subtracting the median helps remove background and stuck features, which brings forward the features that move. **Script:** *do_movie_median.m*
2. Find a good set of parameters for the featuring step. **Script:** *position_finder_singleFrame.m*
3. Run featuring over all of the frames and then track. **Script:** *position_finder_batch.m*
4. Visualise some of the tracks. **Script:** *trajectory_movie.m*
5. Analyse the distribution of feature speeds and velocity orientations. **Script:** *velocity_and_speed_distributions.m*

The aim of our data analysis, besides the training of course, is to compare cell behaviour across experiments.

Potentially interesting questions include (but are not limited to!):

- Does the speed distribution change when the UV light is turned ON? Is there any evidence of the observed phenomenon, and what could be the mechanism?
- Do the speed distributions in the control movies differ? In other words, is there any sort of memory of the previous UV-light exposure history?
- Is there any directionality in cell swimming orientation during photokinesis?

